

# Ngoc-Huynh HO

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## PROFESSIONAL SUMMARY

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AI researcher and postdoctoral fellow with expertise in trustworthy, multimodal, and interpretable machine learning for health-care and human-centered applications. My research focuses on neuroimaging-based dementia prediction, Alzheimer's disease progression modeling, and multimodal representation learning across imaging, clinical, biomarker, and behavioral data. I have a strong record of developing end-to-end AI solutions that combine methodological innovation with practical impact, resulting in peer-reviewed publications, patents, and international collaborations. I am passionate about building fair, generalizable, and clinically meaningful AI systems that improve health outcomes and support real-world decision-making.

## RESEARCH INTERESTS

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Trustworthy AI; multimodal machine learning; neuroimaging and dementia; Alzheimer's disease prediction; disease progression modeling; medical image analysis; self-supervised learning; foundation models for healthcare; explainable and interpretable AI; fairness and bias mitigation in AI; emotion recognition; human-centered AI; biomedical AI.

## SKILLS

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Technical: Machine learning, deep learning, multimodal modeling, medical image analysis, longitudinal prediction, model evaluation, scientific writing, proposal development

Programming: Python, MATLAB, Java, C++/C#, Bash

Tools/Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, MATLAB, Git, Linux/HPC environments

Soft skills: Trustworthy AI, interpretable modeling, cross-disciplinary collaboration, project leadership, mentoring, technical communication

## EDUCATION

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|----------------------------------------------------|---------------|
| <b>PhD in AI Convergence</b>                       | Feb 2021      |
| Chonnam National University, Gwangju, S. Korea     | GPA: 4.25/4.5 |
| <b>MS in Electrical Engineering</b>                | Feb 2017      |
| Kookmin University, Seoul, S. Korea                | GPA: 4.37/4.5 |
| <b>BE in Telecommunications</b>                    | Feb 2015      |
| Ho Chi Minh City University of Technology, Vietnam | GPA: 7.96/10  |

## PROFESSIONAL EXPERIENCE

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| <b>UT Health Science Center at San Antonio, Texas</b>   <i>Instructor</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | June 2026 - Present |
| <ul style="list-style-type: none"><li>Developing trustworthy and fairness-aware AI frameworks for brain age estimation and Alzheimer's disease prediction, focusing on reducing demographic biases and improving generalizability across diverse populations;</li><li>Advancing multimodal learning and foundation models by integrating neuroimaging, clinical, and biomarker data to characterize brain aging, identify early neurodegenerative changes, and improve Alzheimer's disease and related dementia risk prediction.</li></ul> |                     |
| <b>UT Health Science Center at San Antonio, Texas</b>   <i>Postdoctoral Research Fellow</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                | May 2024 - May 2026 |
| <ul style="list-style-type: none"><li>Developing trustworthy AI models for multi-population dementia prediction using advanced few-shot learning techniques;</li><li>Building a multimodal self-supervised foundation model for early dementia prediction, focusing on generalizability, interpretability, and fairness across diverse populations</li></ul>                                                                                                                                                                               |                     |
| <b>Chonnam National University, S. Korea</b>   <i>Post-Doctoral Researcher</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Mar 2021 - Feb 2024 |
| <ul style="list-style-type: none"><li>Developed multimodal deep learning models to predict disease progression (Alzheimer's, Parkinson's, Knee Osteoarthritis);</li><li>Built longitudinal prediction models for Alzheimer's disease using clinical and imaging data;</li><li>Created emotion recognition models using multimodal data (visual, audio, text, neurophysiological signals, personality).</li></ul>                                                                                                                           |                     |
| <b>Chonnam National University, S. Korea</b>   <i>Teaching Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Sep 2021 - Aug 2023 |
| <ul style="list-style-type: none"><li>Guided and evaluated students in Advanced Project for AI Convergence;</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |

- Mentored students on data analysis, machine learning tools, and real-world projects.

**Chonnam National University, S. Korea** | *Research Assistant*

Mar 2020 - Jul 2020

- Research and analyze the applications of AI in enhancing the security of biometrics authentication;
- Researched context-based emotion recognition in the wild.

**University of Oulu, Finland** | *Research Visiting*

Jan 2019

- Collaborated with the Center for Machine Vision and Signal Analysis on emotion recognition trends.

**Chonnam National University, S. Korea** | *Teaching Assistant*

Mar 2018 - Jun 2018

- Conduct necessary exercises of Data Structures course;
- Provide an overall understanding of the data structures as well as an understanding of the classroom process.

**Kookmin University, S. Korea** | *Research Assistant*

Jul 2015 - Dec 2015

- Developed LED-color display and audio jack communication systems using Arduino kits.

## **SELECTED RESEARCH PRESENTATIONS**

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- **NH Ho**. "Toward Equitable AI for Alzheimer's Disease Detection Across Diverse Populations." *Neurepiomics 2026*, San Antonio, TX, USA, October 4–8, 2026. **Invited Talk**.

## **PATENTS**

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1. HJ Yang, **NH Ho**, G Lee, SH Kim. "User emotion prediction system and method using deep learning-based graph fusion". (March 09, 2023). DOI: 10-2023-0034012. CONTRIBUTION: *Proposed, Designed, Implemented, Filed patent*
2. HJ Yang, **NH Ho**, Jahae Kim. "A system and method for predicting progression of Alzheimer's Disease based on a Two-way network". (April 27, 2021). DOI: 10-2022-0147259. CONTRIBUTION: *Proposed, Designed, Implemented, Filed patent*
3. HJ Yang, SH Kim, G Lee, **NH Ho**. "Voice emotion recognition method and system". (February 23, 2021). DOI: 10-2020-0105589. CONTRIBUTION: *Proposed, Designed, Implemented, Filed patent*
4. HJ Yang, SH Kim, ST Jung, SD Joo, **NH Ho**, DS Tran. "System for detecting bone tumor". (January 7, 2021). DOI: 10-2020-0086406. CONTRIBUTION: *Proposed, Designed, Implemented, Filed patent*

## **SOFTWARE REGISTRATION**

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1. HJ Yang, **NH Ho**. "Food recommendations based on real-time sentiment analysis". (October 17, 2023). Registration No.: C-2023-045921. CONTRIBUTION: *Developed, Implemented, Prototyping, Documentation*
2. HJ Yang, **NH Ho**. "Bidirectional progressive-based Alzheimer's disease progression prediction system". (December 24, 2020). Registration No.: C-2020-052306. CONTRIBUTION: *Developed, Implemented, Prototyping, Documentation*
3. HJ Yang, SH Kim, G Lee, **NH Ho**. "Convolutional autoencoder-based program for deep multi-modal clustering". (July 23, 2020). Registration No.: C-2020-024200. CONTRIBUTION: *Developed, Implemented, Prototyping, Documentation*

## **PUBLICATIONS**

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### **Articles in Peer-Reviewed Journals** |

1. S Charisis, NM Pajewski, LR Price, **NH Ho**, T Rashid, D Wang, Y Zeng, NS Kutty, RN Bryan, KC Kern, BC Dickerson, S Seshadri, I Nasrallah, L Launer, C Davatzikos, JD Williamson, M Habes. "Intensive versus standard blood pressure control and overall brain small vessel disease burden: a post-hoc analysis of the SPRINT randomized clinical trial". In: *EClinicalMedicine* 99 (2026), p. 104143. DOI: 10.1016/j.eclinm.2026.104143. CONTRIBUTION: *Review*.
2. **NH Ho**, S Charisis, N Honnorat, SR Brandigampala, D Wang, SR Heckbert, PT Fox, D Martinez, DH Wang, TM Hughes, DB Archer, TJ Hohman, S Seshadri, C Davatzikos, M Habes. "Advancing fair and explainable machine learning for neuroimaging dementia pattern classification in multi-racial and multi-ethnic populations". In: *Nature Communications* 17 (2026), p. 8026. DOI: 10.1038/s41467-026-74515-w. CONTRIBUTION: *Conceptualization, Methodology, Formal Analysis, Visualization, Writing—Original Draft, Writing—Review and Editing*.

3. Ni Honnorat, D Wang, **NH Ho**, D Martinez, SR Brandigampala, SR Heckbert, M Bahrami, JJ Himali, C DeCarli, A Beiser, TM Hughes, S Seshadri, M Habes. "Derivation of machine learning brain aging biomarkers for a set of forty thousand functional connectomes". In: Brain Research Bulletin 237 (2026), p. 111815. DOI: 10.1016/j.brainresbull.2026.111815. CONTRIBUTION: *Review*.
4. **NH Ho**, S Charisis, N Honnorat, SR Brandigampala, D Wang, SR Heckbert, PT Fox, D Martinez, DH Wang, TM Hughes, DB Archer, TJ Hohman, S Seshadri, C Davatzikos, M Habes. "Advancing Fair and Explainable Machine Learning for Neuroimaging Dementia Pattern Classification in Multi-Ethnic Populations". In: bioRxiv (2025), pp. 2025–06. DOI: 10.1101/2025.06.09.658375. CONTRIBUTION: *Conceptualization, Methodology, Formal analysis, Software, Visualization, Writing—original draft, Writing—review & editing*.
5. DP Dao, HJ Yang, J Kim, **NH Ho**. "Longitudinal Alzheimer's Disease Progression Prediction with Modality Uncertainty and Optimization of Information Flow". IEEE Journal of Biomedical and Health Informatics 29 (2025), p. 259–272. DOI: 10.1109/JBHI.2024.3472462. CONTRIBUTION: *Formal analysis, Writing—review & editing*
6. E Lim, **NH Ho**, S Pant, YS Kang, SE Jeon, S Kim, SH Kim, HJ Yang. "GCE: An Audio-Visual Dataset for Group Cohesion and Emotion Analysis". In: Applied Sciences 14 (2024), p. 6742. DOI: 10.3390/app14156742. CONTRIBUTION: *Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing*
7. AQ Duong, **NH Ho**, S Pant, HJ Yang, J Kim. "Residual Relation-Aware Attention Deep Graph-Recurrent Model for Emotion Recognition in Conversation". In: IEEE Access 12 (2024), p. 2349–2360. DOI: 10.1109/ACCESS.2023.3348518. CONTRIBUTION: *Formal analysis, Writing—review & editing*
8. **NH Ho**, HJ Yang, J Kim. "Multimodal multitask learning for predicting MCI to AD conversion using stacked polynomial attention network and adaptive exponential decay". In: Scientific Reports 13 (2023), p. 11243. DOI: 10.1038/s41598-023-37500-7. CONTRIBUTION: *Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing*
9. TD Tran, **NH Ho**, S Pant, HJ Yang, SH Kim, G Lee. "Cross-modality learning by exploring modality interactions for emotion reasoning". In: IEEE Access 11 (2023), p. 56634–56648. DOI: 10.1109/ACCESS.2023.3283597. CONTRIBUTION: *Formal analysis, Visualization, Writing—review & editing*
10. **NH Ho**, HJ Yang, J Kim, DP Dao, HR Park, S Pant. "Predicting progression of Alzheimer's disease using forward-to-backward bi-directional network with integrative imputation". In: Neural Networks 150 (2022), p. 422–439. DOI: 10.1016/j.neunet.2022.03.016. CONTRIBUTION: *Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing*
11. **NH Ho**, HJ Yang, SH Kim, G Lee, SB Yoo. "Deep Graph Fusion based Multimodal Evoked Expressions from Large-Scale Videos". In: IEEE Access 9 (2021). DOI: 10.1109/ACCESS.2021.3107548. CONTRIBUTION: *Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing*
12. TD Tran, J Kim, **NH Ho**, HJ Yang, S Pant, SH Kim, G Lee. "Stress Analysis with Dimensions of Valence and Arousal in the Wild". In: Applied Sciences 11.11 (2021), p. 5194. DOI: 10.3390/app11115194. CONTRIBUTION: *Writing—review & editing*
13. DS Tran, **NH Ho**, HJ Yang, SH Kim, G Lee. "Real-time virtual mouse system using RGB-D images and fingertip detection". In: Multimedia Tools and Applications 80 (2021), p. 10473–10490. DOI: 10.1007/s11042-020-10156-5. CONTRIBUTION: *Writing—review & editing*
14. **NH Ho**, HJ Yang, SH Kim, G Lee. "Multimodal approach of speech emotion recognition using multi-level multi-head fusion attention-based recurrent neural network". In: IEEE Access 8 (2020), p. 61672–61686. DOI: 10.1109/ACCESS.2020.2984368. CONTRIBUTION: *Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing*

15. TLB Khanh, DP Dao, **NH Ho**, HJ Yang, ET Baek, G Lee, SH Kim, SB Yoo. "Enhancing U-Net with spatial-channel attention gate for abnormal tissue segmentation in medical imaging". In: Applied Sciences 10.17 (2020), p. 5729. DOI: 10.3390/app10175729. **CONTRIBUTION: Formal analysis, Writing—review & editing**
16. DS Tran, **NH Ho**, HJ Yang, ET Baek, SH Kim, G Lee. "Real-time hand gesture spotting and recognition using RGB-D camera and 3D convolutional neural network". In: Applied Sciences 10.2 (2020), p. 722. DOI: 10.3390/app10020722. **CONTRIBUTION: Formal analysis, Writing—review & editing**
17. **NH Ho**, HJ Yang, SH Kim, ST Jung, SD Joo. "Regenerative semi-supervised bidirectional W-network-based knee bone tumor classification on radiographs guided by three-region bone segmentation". In: IEEE Access 7 (2019), p. 6825-6833. DOI: 110.1109/ACCESS.2019.2949125. **CONTRIBUTION: Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing**
18. TD Vu, **NH Ho**, HJ Yang, J Kim, HC Song. "Non-white matter tissue extraction and deep convolutional neural network for Alzheimer's disease detection". In: Soft Computing 22 (2018), p. 61672-61686. DOI: 10.1007/s00500-018-3421-5. **CONTRIBUTION: Conceptualization, Formal analysis, Methodology, Visualization, Writing—review & editing**
19. PH Truong, ND Nguyen, **NH Ho**, GM Jeong. "Nonparametric regression-based step-length estimation for arm-swing walking using a smartphone". In: IJCCC 13.4 (2018), p. 566-573. DOI: 10.15837/ijccc.2018.4.3148. **CONTRIBUTION: Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft**
20. **NH Ho**, PH Truong, GM Jeong. "Step-detection and adaptive step-length estimation for pedestrian dead-reckoning at various walking speeds using a smartphone". In: Sensors 16.9 (2016), p. 1423. DOI: 10.3390/s16091423. **CONTRIBUTION: Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing**
21. VK Ho, NK Doan, **NH Ho**. "Impact of channel estimation error on the performance of relay selection in cognitive radio networks". In: Wireless Personal Communications 84 (2015), p. 2513-2536. DOI: 10.1007/s11277-015-2717-3. **CONTRIBUTION: Conceptualization, Formal analysis, Methodology, Visualization, Writing—original draft, Writing—review & editing**

#### Articles in International Conferences/Proceedings |

1. **NH Ho**, S Charisis, D Wang, SR Heckbert, DM Martinez, TM Hughes, N Honnorat, DB Archer, TJ Hohman, S Seshadri, C Davatzikos, M Habes. "Addressing AI Bias in MRI-Based Dementia Classification Among Hispanic Individuals". In: The Alzheimer's Association International Conference. (AAIC'25): Toronto, Canada. July, 2025
2. **NH Ho**, HJ Yang, J Kim. "Prediction of Alzheimer's Disease Progression using Multiview Dense Residual Attention and Stack Polynomial Attention". In: The 11th International Conference on Big Data Applications and Services. (BigDAS'23): Danang, Vietnam. August 16, 2023
3. DP Dao, HJ Yang, G Lee, SH Kim, **NH Ho**, S Pant, SR Kang, IJ Oh. "Survival Analysis based on Lung Tumor Segmentation using Global Context-aware Transformer in Multimodality". In: The 26th International Conference on Pattern Recognition. (ICPR'22): Montreal Quebec, Canada. August 21, 2022
4. DK Nguyen, S Pant, **NH Ho**, G Lee, SH Kim, HJ Yang. "Affective Behavior Analysis using Action Unit Relation Graph and Multi-task Cross Attention". In: The European Conference on Computer Vision Workshops. (ECCV'22): Tel Aviv, Israel. October 23, 2022
5. AQ Duong, **NH Ho**, HJ Yang, G Lee, SH Kim. "Multi-modal Stress Recognition Using Temporal Convolution and Recurrent Network with Positional Embedding". In: The 2nd Multimodal Sentiment Analysis Challenge (MuSe'21): Chengdu, China, October 24, 2021. ACM Multimedia 2021.
6. **NH Ho**, HJ Yang, J Kim, DP Dao, S Pant. "RASurv: Residual Attention-aware Method for Progression-free Survival of Alzheimer's Disease". In: The 5th International Conference on Big Data, IoT, and Cloud Computing (BIC'21): Jeju Island, S. Korea, August 16-18, 2021.

7. HD Le, HJ Yang, **NH Ho**, S Pant. "Attention-based Image and Text Fusion for Deep Multimodal Classification in Disaster Analysis". In: The 5th International Conference on Big Data, IoT, and Cloud Computing (BIC'21): Jeju Island, S. Korea, August 16-18, 2021.
8. DP Dao, **NH Ho**, J Kim, HJ Yang. "Improving Recurrent Gate Mechanism For Time-to-Conversion Prediction Of Alzheimer's Disease". In: The 9th International Conference on Smart Media and Applications (SMA'20): Jeju Island, S. Korea, September 17-19, 2020.
9. **NH Ho**, HJ Yang, HC Song, J Kim. "ADeepTool: Application Tool for Alzheimer's Disease Diagnosis Based on Deep Learning Approach". In: The 7th International Conference on Big Data Applications and Services (BigDAS'19): Jeju Island, S. Korea, August 21-24, 2019.
10. **NH Ho**, HJ Yang, LN Do, SH Kim, G Lee. "Early Fusion-based Emotion Recognition Using Multiple Audio Features from GMM Supervector and Deep Convolutional Neural Network". In: The 6th International Conference on Big Data Applications and Services (BigDAS'18): Zhengzhou, China, August 19-22, 2018.
11. DS Tran, HJ Yang, SH Kim, GS Lee, LN Do, **NH Ho**, VQ Nguyen. "Audio-based emotion recognition using GMM Supervector an SVM linear kernel". In: The 2nd International Conference on Machine Learning and Soft Computing (ICMLSC'18): Phu Quoc Island, Vietnam, February 2018.
12. LN Do, HJ Yang, **NH Ho**, SH Kim, G Lee. "A Fusion Model for Emotion Recognition from Audio- Video Data". In: The 5th International Conference on Big Data Applications and Services (BigDAS'17): Jeju Island, S. Korea, November 23-25, 2017.
13. TD Vu, **NH Ho**, JM Joo, SH Kim, YC Kim, HJ Yang, J Kim, HC Song. "Detect Alzheimer's disease by Multimodal Deep Learning Network using Convolutional Autoencoder". In: The 1st International Conference on Big Data, IoT, and Cloud Computing (BIC'17): Jeju Island, S. Korea, August 22-24, 2017.

#### Articles in Korean Domestic Journals/Conferences/Proceedings |

1. TC Do, HJ Yang, **NH Ho**. "Application of Deep Recurrent Q Network with Dueling Architecture for Optimal Sepsis Treatment Policy". In: Smart Media Journals 10 (2021). 10.30693/SMJ.2021.10.2.48.
2. AQ Duong, **NH Ho**, HJ Yang, SH Kim, G Lee, AR Oh. "Multimodal Deep Graph Fusion for Evoked Expression Recognition in Large-Scale Videos". In: 2021 Conference of Korea Computer Congress (KCC'21): Jeju Island, S. Korea, June 23-25, 2021.
3. **NH Ho**, HJ Yang, J Kim, H Jeong. "A Comparative Study for Estimation of Time to Alzheimer's Disease based on Machine Learning Regressors". In: 2020 Spring Conference of Korean Institute of Smart Media (KISM'20): Gwangju, S. Korea, May 22-23, 2020.
4. **NH Ho**, HJ Yang, SH Kim, G Lee. "Group-based Cohesion Prediction using Multi-Task Learning and Confidence of Adaptive Ranging". In: 2019 August Conference of Korean Institute of Smart Media (KISM'19): Gwangju, S. Korea, November 07-08, 2019.
5. **NH Ho**, HJ Yang, SH Kim, G Lee. "Textual Emotion Recognition Based on Recurrent Neural Network for Conscious Conservation". In: 2019 Spring Conference of Korean Institute of Smart Media (KISM'19): Chungju, S. Korea, April 26-27, 2019.
6. **NH Ho**, HJ Yang, SH Kim, ST Jeong, SD Joo. "End-to-end Knee Bone Tumor Classification from Radiographs Based on Semi-Supervised Ensemble Wnet". In: 2018 August Conference of Korean Multimedia Society (KMS'18): Gwangju, S. Korea, October 20, 2018.
7. **NH Ho**, DS Tran, JM Joo, HJ Yang, SH Kim, ST Jung, SD Joo. "Bone Area Segmentation in X-Ray images using an Adaptable U-Net". In: 2018 Conference of Korea Computer Congress (KCC'18): Jeju Island, S. Korea, June 20-22, 2018.

**REFERENCES**

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Available upon request.